

Q1: Color Scales for Data Representation [R]

R program (questions 1 and 2)

```
library(ggplot2)

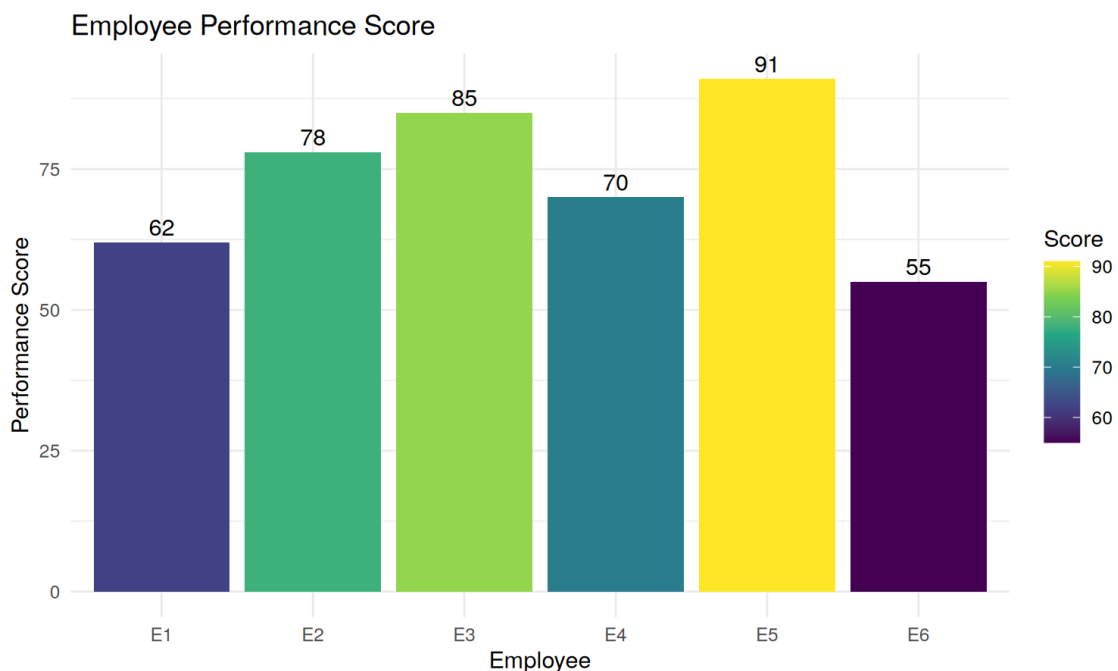
# Data
emp <- data.frame(
  Employee = c("E1", "E2", "E3", "E4", "E5", "E6"),
  Score    = c(62, 78, 85, 70, 91, 55)
)

# Bar chart with a sequential, colour-blind-safe scale (viridis: dark -> bright = low -> high)
p <- ggplot(emp, aes(x = Employee, y = Score, fill = Score)) +
  geom_col() +
  geom_text(aes(label = Score), vjust = -0.4) +
  scale_fill_viridis_c(option = "D", direction = 1, name = "Score") +
  labs(title = "Employee Performance Score", x = "Employee", y = "Performance Score") +
  theme_minimal()
print(p)

# Highest and lowest performers
emp[which.max(emp$Score), ]
emp[which.min(emp$Score), ]
```

Output

```
Employee Score
5      E5    91
Employee Score
6      E6    55
```



1. Write an R program to create a bar chart using a sequential color scale based on Performance Score.

Done: the program above draws one bar per employee and maps the Performance Score to the fill colour with a sequential scale (`scale_fill_viridis_c`), so a darker/brighter shade always means a higher score.

2. Identify the employees with the highest and lowest performance scores from the visualization.

Highest: E5 with 91 (the brightest bar). Lowest: E6 with 55 (the darkest bar). E3 (85) is the next best performer.

3. Explain why a sequential color scale is appropriate.

Performance Score is an ordered, numeric variable that runs from low to high with no meaningful midpoint. A sequential scale uses one steady progression of lightness, so the colour order matches the value order and viewers can tell at a glance which employees score higher or lower. A diverging scale would wrongly suggest a neutral centre, and a categorical scale would give no sense of order at all.

4. Suggest a color palette suitable for color-blind users.

The viridis palette (used in the program) is designed for color-blind viewers: it changes steadily in lightness from dark purple to yellow, so values stay distinguishable under red-green colour blindness. Other safe choices are ColorBrewer's single-hue Blues or YlGnBu (as long as the lightest shade is still visible on a white background). Avoid red-green combinations.

5. Explain the difference between sequential and categorical color scales.

A sequential scale is for ordered or numeric data: one hue (or a smooth path of hues) that goes from light to dark, so colour intensity shows magnitude. A categorical (qualitative) scale is for nominal groups that have no order: clearly different hues, so colour only says 'different group' and not 'more' or 'less'.

Q2: Legends and Annotations [R]

R program (questions 1 and 2)

```
library(ggplot2)

# Data
stock <- data.frame(
  Month = factor(rep(c("Jan", "Feb", "Mar", "Apr", "May", "Jun"), 3),
    levels = c("Jan", "Feb", "Mar", "Apr", "May", "Jun")),
  Company = rep(c("Co. X", "Co. Y", "Co. Z"), each = 6),
  Price = c(410, 430, 460, 480, 520, 500,
    380, 395, 410, 430, 460, 490,
    450, 460, 440, 470, 510, 530)
)

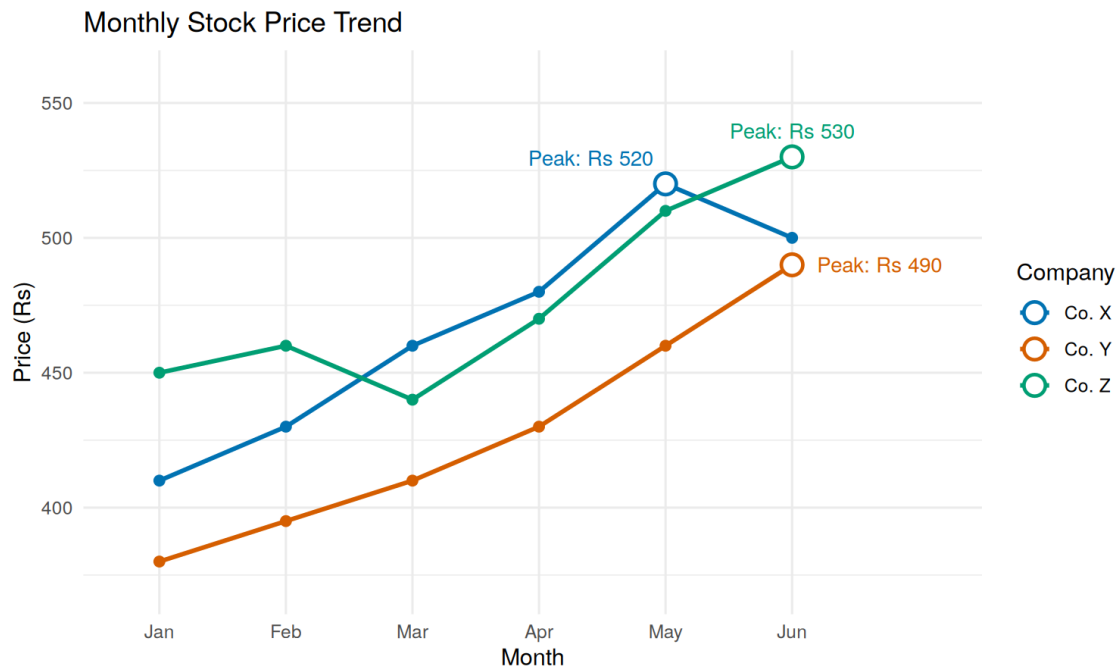
# Row of highest price for each company
peak <- do.call(rbind, lapply(split(stock, stock$Company),
  function(d) d[which.max(d$Price), ]))

print(peak)
peak$vj <- c(-1.2, 0.5, -1.2) # place each label where it does not sit on a line
peak$hj <- c(1.1, -0.2, 0.5)

# Multi-series line chart with legend and annotations at the peaks
p <- ggplot(stock, aes(x = Month, y = Price, colour = Company, group = Company)) +
  geom_line(linewidth = 1) +
  geom_point(size = 2) +
  geom_point(data = peak, size = 4, shape = 21, fill = "white", stroke = 1.2) +
  geom_text(data = peak, aes(label = paste0("Peak: Rs ", Price), vjust = vj, hjust = hj),
    show.legend = FALSE, size = 3.5) +
  scale_colour_manual(values = c("Co. X" = "#0072B2", "Co. Y" = "#D55E00", "Co. Z" = "#009E73")) +
  scale_x_discrete(expand = expansion(add = c(0.6, 1.5))) +
  scale_y_continuous(limits = c(370, 560)) +
  labs(title = "Monthly Stock Price Trend", x = "Month", y = "Price (Rs)", colour = "Company") +
  theme_minimal() + theme(legend.position = "right")
print(p)
```

Output (highest price row for each company)

	Month	Company	Price
Co. X	May	Co. X	520
Co. Y	Jun	Co. Y	490
Co. Z	Jun	Co. Z	530



1. Write an R program to plot the monthly stock-price trend for all three companies on a single chart with a clear legend.

Done: the data is put in long format (Month, Company, Price) and plotted with `geom_line()` and `geom_point()`, with colour mapped to Company so ggplot2 creates a legend on the right (titled 'Company') and uses colour-blind-safe colours.

2. Annotate (using `annotate()/geom_text()`) the point of highest price for each company.

Co. X peaks in May at Rs 520, Co. Y peaks in June at Rs 490, and Co. Z peaks in June at Rs 530. Each peak is circled and labelled with `geom_text()`, and the labels are placed so they do not sit on any line.

3. Explain the role of legends in multi-series visualizations.

A legend links each colour (or line style/shape) to the series it represents. With three lines on one chart, the viewer cannot tell which line belongs to Co. X, Co. Y or Co. Z without it, so the legend turns the picture into readable information.

4. Discuss best practices for placing legends to avoid obscuring data.

Place the legend outside the plotting area (right or bottom) rather than on top of lines and points; keep labels short; keep the legend order the same as the order of the lines; use the same colours in the legend and the plot; and if a legend must be inside the plot, put it only in an empty corner. In ggplot2 this is controlled with `theme(legend.position = ...)`.

5. Suggest an alternative to a legend for distinguishing categories (e.g., direct labeling).

Direct labeling: write each company's name at the end of its own line (using `geom_text()` or `geom_label()`, or the `ggrepel/directlabels` packages) and remove the legend. The viewer reads the label right next to the line, so there is no back-and-forth between legend and chart.

Q3: Mapping Data onto Aesthetics [R]

R program (questions 1 and 2)

```
library(ggplot2)

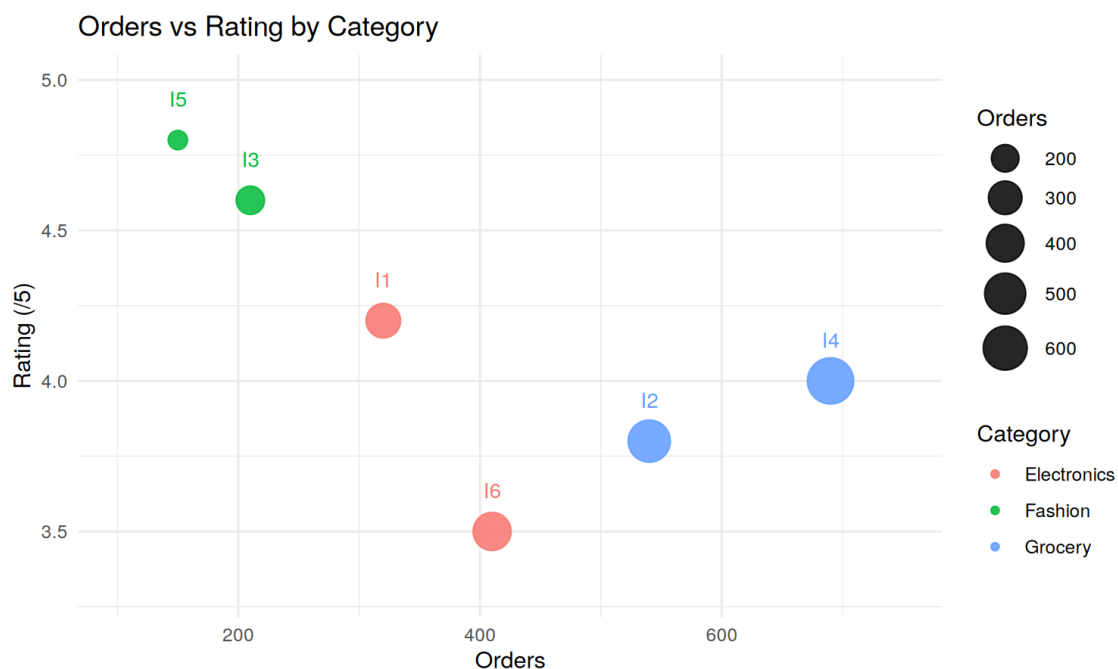
# Data
items <- data.frame(
  Item      = c("I1", "I2", "I3", "I4", "I5", "I6"),
  Orders    = c(320, 540, 210, 690, 150, 410),
  Rating    = c(4.2, 3.8, 4.6, 4.0, 4.8, 3.5),
  Category  = c("Electronics", "Grocery", "Fashion", "Grocery", "Fashion", "Electronics")
)

# Scatter plot: Orders (x) vs Rating (y), colour = Category, size = Orders
p <- ggplot(items, aes(x = Orders, y = Rating, colour = Category, size = Orders)) +
  geom_point(alpha = 0.85) +
  geom_text(aes(label = Item), vjust = -2.2, size = 3.5, show.legend = FALSE) +
  scale_size_continuous(range = c(4, 10)) +
  scale_x_continuous(limits = c(100, 750)) +
  scale_y_continuous(limits = c(3.3, 5.0)) +
  labs(title = "Orders vs Rating by Category", x = "Orders", y = "Rating (/5)") +
  theme_minimal()
print(p)

# Strength and direction of the relationship
cor(items$Orders, items$Rating)
```

Output (correlation between Orders and Rating)

```
[1] -0.7044823
```



1. Write an R program (using ggplot2) to create a scatter plot with Orders on the x-axis and Rating on the y-axis.

Done: `ggplot(items, aes(x = Orders, y = Rating))` with `geom_point()`, as shown in the program above.

2. Map Category using an appropriate aesthetic (e.g. colour) and Orders using another suitable aesthetic (e.g. size).

Category is mapped to colour (a categorical variable suits distinct hues) and Orders is mapped to point size (a numeric variable suits size), using `aes(colour = Category, size = Orders)`. Orders is shown on both the x-axis and the size, which makes high-order items stand out.

3. Interpret the relationship between Orders and Rating.

There is a moderate negative relationship (correlation about -0.70): items with more orders tend to have lower ratings. For example, I4 has the most orders (690) with a rating of 4.0, while I5 has the fewest (150) and the highest rating (4.8). With only six items this is only a tendency, and it does not prove that more orders cause lower ratings.

4. Identify the type of scale used for each variable (nominal, ordinal, continuous).

Item: nominal (an identifier). Orders: continuous (numeric count on a ratio scale). Rating (/5): continuous (a numeric score on a 0-5 scale; if it were individual star ratings it would be ordinal). Category: nominal.

5. Suggest one alternative visualization that better represents the same data.

A bar chart of Orders for each Item, filled by Category, with the Rating written on or above each bar (or shown in a second panel). With only six distinct items, bars make each item easy to read, and the Category colour and Rating can be shown without relying on point size.

Q4: Correlation Heatmap [R]

R program (question 1)

```
library(ggplot2)
library(reshape2)

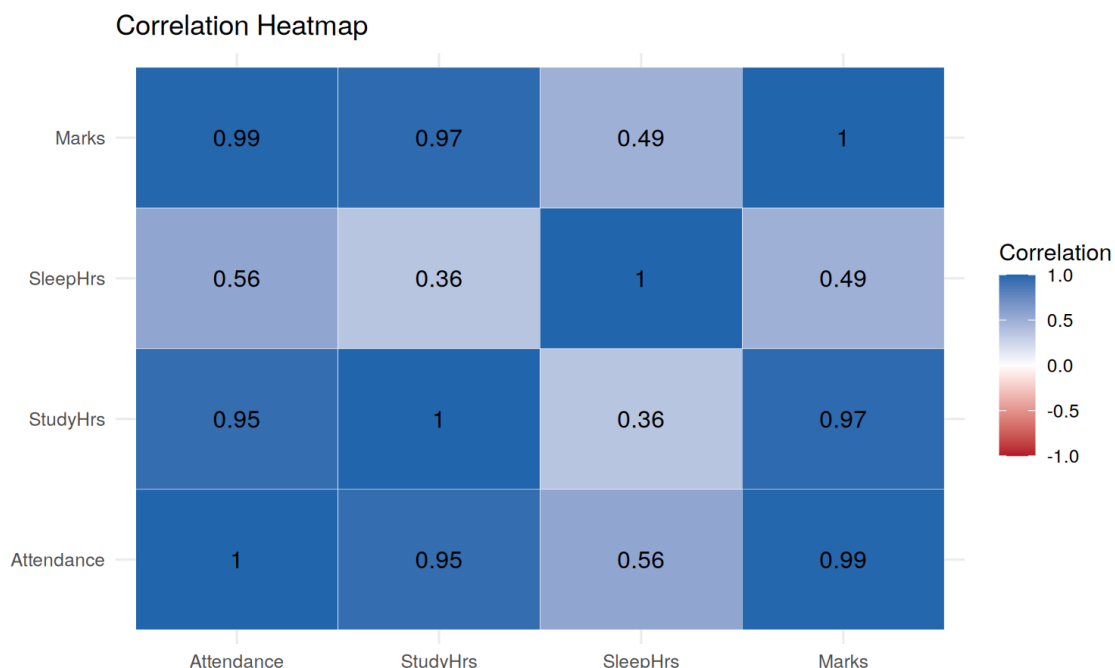
# Data
df <- data.frame(
  Attendance = c(92, 75, 88, 60, 95, 70),
  StudyHrs   = c(3.5, 1.5, 2.8, 1.0, 4.0, 2.0),
  SleepHrs   = c(6.5, 8.0, 7.0, 5.5, 7.5, 6.0),
  Marks      = c(88, 62, 80, 48, 91, 58)
)

# Correlation matrix
cm <- round(cor(df), 2)
print(cm)

# Heatmap with a diverging colour scale centred at 0
long <- melt(cm)
p <- ggplot(long, aes(x = Var1, y = Var2, fill = value)) +
  geom_tile(colour = "white") +
  geom_text(aes(label = value), size = 4) +
  scale_fill_gradient2(low = "#B2182B", mid = "white", high = "#2166AC",
    midpoint = 0, limits = c(-1, 1), name = "Correlation") +
  labs(title = "Correlation Heatmap", x = NULL, y = NULL) +
  theme_minimal()
print(p)
```

Output (correlation matrix)

	Attendance	StudyHrs	SleepHrs	Marks
Attendance	1.00	0.95	0.56	0.99
StudyHrs	0.95	1.00	0.36	0.97
SleepHrs	0.56	0.36	1.00	0.49
Marks	0.99	0.97	0.49	1.00



1. Write an R program to compute the correlation matrix of Attendance, Study Hours, Sleep Hours, and Marks, and display it as a heatmap.

Done: cor() computes the matrix, melt() converts it to long format, and geom_tile() draws the heatmap with the correlation values printed in each cell.

2. Identify the pair of variables with the strongest correlation.

Attendance and Marks, with $r = 0.99$. The next strongest are Study Hours and Marks (0.97) and Attendance and Study Hours (0.95).

3. Explain what a diverging color scale communicates in a correlation heatmap.

A diverging scale uses two different hues that meet at a neutral colour (white) at the midpoint 0: here blue for positive correlation and red for negative correlation. The colour shows the direction of the relationship, and the intensity shows its strength, with white meaning no correlation. In this data all correlations are positive, so every cell is blue.

4. Discuss one limitation of using correlation to interpret this data.

Correlation only measures how two variables move together; it does not show cause and effect. For example, high attendance and high marks may both be driven by a student's motivation. Also, with only six students the values are unstable, and a single unusual student could change them a lot.

5. Suggest one additional variable that could improve this analysis.

Each student's previous-semester marks (or CGPA). It captures prior ability, so the analysis can separate the effect of attendance and study time from how well the student was already performing.

Q5: Stacked Bar Charts and Proportions [R]

R program (questions 1 and 2)

```
library(ggplot2)
library(reshape2)

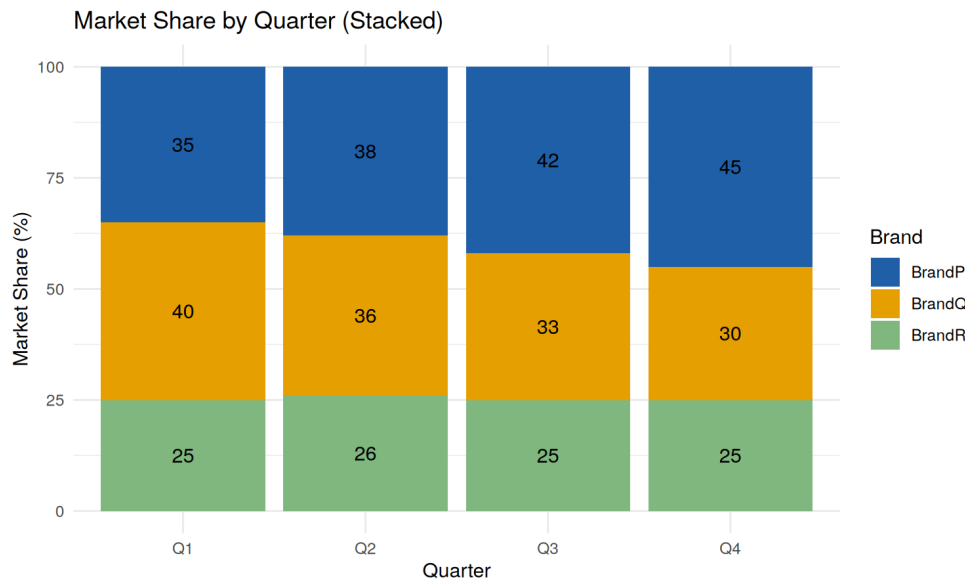
# Data
share <- data.frame(
  Quarter = c("Q1", "Q2", "Q3", "Q4"),
  BrandP = c(35, 38, 42, 45),
  BrandQ = c(40, 36, 33, 30),
  BrandR = c(25, 26, 25, 25)
)
long <- melt(share, id.vars = "Quarter", variable.name = "Brand", value.name = "Share")

cols <- c(BrandP = "#1F5FA8", BrandQ = "#E69F00", BrandR = "#7FB77E")

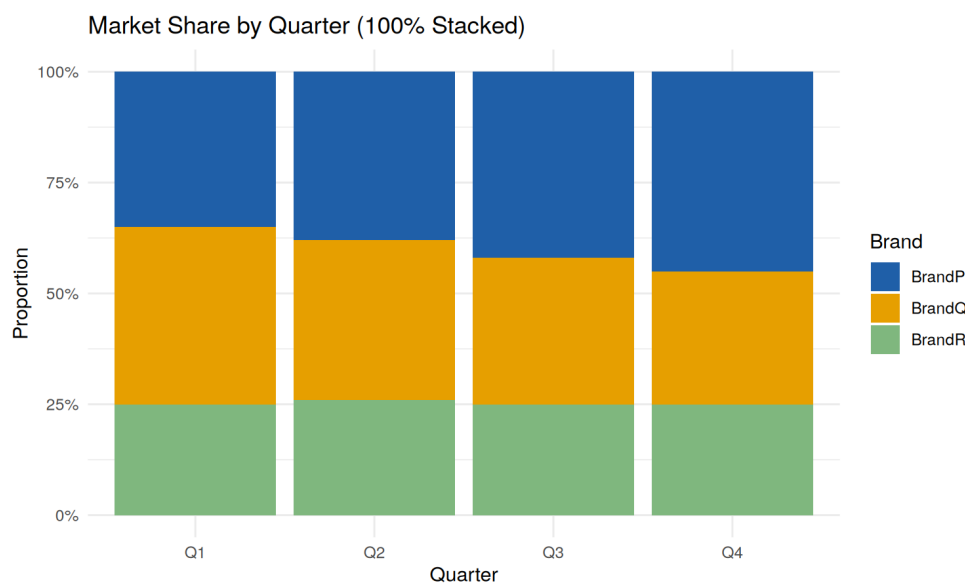
# (1) Stacked bar chart of market share by quarter
p1 <- ggplot(long, aes(x = Quarter, y = Share, fill = Brand)) +
  geom_col() +
  geom_text(aes(label = Share), position = position_stack(vjust = 0.5)) +
  scale_fill_manual(values = cols) +
  labs(title = "Market Share by Quarter (Stacked)", y = "Market Share (%)") +
  theme_minimal()
print(p1)

# (2) 100% (proportional) stacked bar chart
p2 <- ggplot(long, aes(x = Quarter, y = Share, fill = Brand)) +
  geom_col(position = "fill") +
  scale_y_continuous(labels = scales::percent) +
  scale_fill_manual(values = cols) +
  labs(title = "Market Share by Quarter (100% Stacked)", y = "Proportion") +
  theme_minimal()
print(p2)
```

Output: stacked bar chart



Output: 100% (proportional) stacked bar chart



1. Write an R program to create a stacked bar chart showing each brand's market share (%) by Quarter.

Done: the data is melted to long format and plotted with `geom_col()`, filling by Brand, with each segment labelled with its percentage.

2. Modify the program to create a 100% (proportional) stacked bar chart.

Done: change `geom_col()` to `geom_col(position = "fill")` and format the y-axis as percentages with `scale_y_continuous(labels = scales::percent)`, as in the second plot.

3. Compare what each version communicates about the brands' relative performance.

The stacked chart shows each brand's actual share value: Brand P grows from 35% to 45%, Brand Q falls from 40% to 30%, and Brand R stays at about 25%. The 100% chart shows each brand's proportion of the quarter's total. Because the quarterly shares here already add up to 100%, the two charts look identical for this data. They would differ only if the totals varied between quarters.

4. Identify the brand with a consistently rising trend.

Brand P: 35 -> 38 -> 42 -> 45, rising in every quarter. Brand Q falls every quarter, and Brand R stays almost flat.

5. Explain a limitation of stacked bar charts when comparing individual segment sizes.

Only the bottom segment starts from a common baseline. The other segments start at different heights, so their lengths are hard to compare by eye across bars. For example, it is difficult to judge exactly how Brand Q changed from quarter to quarter. A grouped bar chart or a line chart is better for comparing individual segments.